

The Notebooks · Era II — Operate

Gears / early mechanism

Testing Mode · UK · IB · US

Exercises

T.II.1.001 · Roman numerals review in large-number context · core · UK_KS2

- (a) [never/neutral/1mk] Convert 66 to Roman numerals (skill: Roman numerals review in large-number context).
- (b) [never/UK/1mk] What Arabic number is LXXX?
- (c) [never/UK/2mk] A page is numbered LXVI. Write the next page number in Roman numerals.

T.II.1.002 · Formal column addition multi-digit exam pack · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $762 + 350$.
- (b) [never/UK/2mk] Work out $772 + 345$.
- (c) [never/UK/2mk] Score 762 then 350 more. Total?

T.II.1.003 · Formal column subtraction multi-digit exam pack · core · UK_KS2

- (a) [never/neutral/2mk] Column subtraction: $755 - 377$.
- (b) [never/UK/2mk] Calculate $752 - 379$.
- (c) [never/UK/2mk] A library has 755 books; 377 are loaned. How many remain?

T.II.1.004 · Round to estimate before calculating (UK) · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $438 + 425$.
- (b) [never/UK/2mk] Work out $448 + 420$.
- (c) [never/UK/2mk] Score 438 then 425 more. Total?

T.II.1.005 · Numbers to 10 million reading/writing · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $574 + 380$.
- (b) [never/UK/2mk] Work out $584 + 375$.
- (c) [never/UK/2mk] Score 574 then 380 more. Total?

T.II.1.006 · Negative numbers on number lines (Y5) · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $301 + 505$.
- (b) [never/UK/2mk] Work out $311 + 500$.
- (c) [never/UK/2mk] Score 301 then 505 more. Total?

T.II.1.007 · Count forwards/backwards in powers of 10 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $691 + 174$.
- (b) [never/UK/2mk] Work out $701 + 169$.
- (c) [never/UK/2mk] Score 691 then 174 more. Total?

T.II.1.008 · Compare numbers using place-value charts · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $735 + 259$.
- (b) [never/UK/2mk] Work out $745 + 254$.
- (c) [never/UK/2mk] Score 735 then 259 more. Total?

T.II.1.009 · Formal column methods mixed 1 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $617 + 623$.
- (b) [never/UK/2mk] Work out $627 + 618$.
- (c) [never/UK/2mk] Score 617 then 623 more. Total?

T.II.1.010 · Formal column methods mixed 2 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $426 + 391$.
- (b) [never/UK/2mk] Work out $436 + 386$.
- (c) [never/UK/2mk] Score 426 then 391 more. Total?

T.II.1.011 · Formal column methods mixed 3 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $810 + 597$.
- (b) [never/UK/2mk] Work out $820 + 592$.
- (c) [never/UK/2mk] Score 810 then 597 more. Total?

T.II.2.001 · 12×12 tables fluency milestone pack · core · UK_KS2

- (a) [never/neutral/2mk] Calculate 48×28 (formal method).
- (b) [never/UK/1mk] What is 48×10 ?
- (c) [never/UK/2mk] A pack has 48 stickers; buy 28 packs. Total stickers?

T.II.2.002 · Long multiplication (formal written method) · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $573 + 391$.
- (b) [never/UK/2mk] Work out $583 + 386$.
- (c) [never/UK/2mk] Score 573 then 391 more. Total?

T.II.2.003 · Grid / area-model to column bridge · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/2mk] Calculate 47×25 (formal method).
- (b) [never/UK/1mk] What is 47×10 ?
- (c) [never/UK/2mk] A pack has 47 stickers; buy 25 packs. Total stickers?

T.II.2.004 · Multiplication worded multi-step SATs · challenge · UK_KS2

- (a) [never/neutral/2mk] Calculate 14×16 (formal method).
- (b) [never/UK/1mk] What is 14×10 ?
- (c) [never/UK/2mk] A pack has 14 stickers; buy 16 packs. Total stickers?

T.II.2.005 · Multiply by 10, 100, 1000 place-value moves · core · UK_KS2

- (a) [never/neutral/2mk] Calculate 41×29 (formal method).
- (b) [never/UK/1mk] What is 41×10 ?
- (c) [never/UK/2mk] A pack has 41 stickers; buy 29 packs. Total stickers?

T.II.2.006 · Square numbers visual arrays pack · core · UK_KS2

- (a) [never/neutral/2mk] Calculate 42×15 (formal method).
- (b) [never/UK/1mk] What is 42×10 ?
- (c) [never/UK/2mk] A pack has 42 stickers; buy 15 packs. Total stickers?

T.II.2.007 · Distributive property for mental mult · core · UK_KS2

- (a) [never/neutral/2mk] Calculate 47×22 (formal method).
- (b) [never/UK/1mk] What is 47×10 ?
- (c) [never/UK/2mk] A pack has 47 stickers; buy 22 packs. Total stickers?

T.II.2.008 · Multiples vs factors language traps · core · UK_KS2

- (a) [never/neutral/2mk] Calculate 45×15 (formal method).
- (b) [never/UK/1mk] What is 45×10 ?
- (c) [never/UK/2mk] A pack has 45 stickers; buy 15 packs. Total stickers?

T.II.2.009 · Long multiplication accuracy set 1 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $287 + 317$.
- (b) [never/UK/2mk] Work out $297 + 312$.
- (c) [never/UK/2mk] Score 287 then 317 more. Total?

T.II.2.010 · Long multiplication accuracy set 2 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $465 + 257$.
- (b) [never/UK/2mk] Work out $475 + 252$.
- (c) [never/UK/2mk] Score 465 then 257 more. Total?

T.II.2.011 · Long multiplication accuracy set 3 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $303 + 442$.
- (b) [never/UK/2mk] Work out $313 + 437$.
- (c) [never/UK/2mk] Score 303 then 442 more. Total?

T.II.3.001 · Long division formal method (one-digit divisor) · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $505 + 350$.
- (b) [never/UK/2mk] Work out $515 + 345$.
- (c) [never/UK/2mk] Score 505 then 350 more. Total?

T.II.3.002 · Long division by two-digit divisors · challenge · UK_KS2

- (a) [never/neutral/2mk] Column addition: $284 + 583$.
- (b) [never/UK/2mk] Work out $294 + 578$.
- (c) [never/UK/2mk] Score 284 then 583 more. Total?

T.II.3.003 · Division with remainders as fractions/decimals · core · UK_KS2

- (a) [never/neutral/1mk] Divide $135 \div 3$.
- (b) [never/UK/2mk] Using short division: $137 \div 3$. Give quotient and remainder.
- (c) [never/UK/2mk] 137 sweets shared among 3 children. How many each and remainder?

T.II.3.004 · Short division (bus-stop) method fluency · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $385 + 263$.
- (b) [never/UK/2mk] Work out $395 + 258$.
- (c) [never/UK/2mk] Score 385 then 263 more. Total?

T.II.3.005 · Division worded multi-step SATs · challenge · UK_KS2

- (a) [never/neutral/1mk] Divide $312 \div 8$.
- (b) [never/UK/2mk] Using short division: $319 \div 8$. Give quotient and remainder.
- (c) [never/UK/2mk] 319 sweets shared among 8 children. How many each and remainder?

T.II.3.006 · Chunking division as bridge method · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $826 + 395$.
- (b) [never/UK/2mk] Work out $836 + 390$.
- (c) [never/UK/2mk] Score 826 then 395 more. Total?

T.II.3.007 · Worded division sharing vs grouping · core · UK_KS2

- (a) [never/neutral/1mk] Divide $369 \div 9$.
- (b) [never/UK/2mk] Using short division: $376 \div 9$. Give quotient and remainder.
- (c) [never/UK/2mk] 376 sweets shared among 9 children. How many each and remainder?

T.II.3.008 · Long division accuracy set 1 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $600 + 247$.
- (b) [never/UK/2mk] Work out $610 + 242$.
- (c) [never/UK/2mk] Score 600 then 247 more. Total?

T.II.3.009 · Long division accuracy set 2 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $769 + 386$.
- (b) [never/UK/2mk] Work out $779 + 381$.
- (c) [never/UK/2mk] Score 769 then 386 more. Total?

T.II.3.010 · Long division accuracy set 3 · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $381 + 269$.
- (b) [never/UK/2mk] Work out $391 + 264$.
- (c) [never/UK/2mk] Score 381 then 269 more. Total?

T.II.4.001 · Prime numbers to 100 recognition drills · core · UK_KS2,UK_KS3

- (a) [never/neutral/1mk] List all factors of 20.
- (b) [never/UK/1mk] HCF(20,30)?
- (c) [never/UK/2mk] LCM(20,30)?

T.II.4.002 · HCF and LCM worded problems (UK) · core · UK_KS2,UK_KS3

- (a) [never/neutral/1mk] List all factors of 14.
- (b) [never/UK/1mk] HCF(14,21)?
- (c) [never/UK/2mk] LCM(14,21)?

T.II.4.003 · Square and cube numbers recall pack · core · UK_KS2

- (a) [never/neutral/1mk] List all factors of 16.
- (b) [never/UK/1mk] HCF(16,24)?
- (c) [never/UK/2mk] LCM(16,24)?

T.II.4.004 · Common factors and common multiples lists · core · UK_KS2

- (a) [never/neutral/1mk] List all factors of 14.
- (b) [never/UK/1mk] HCF(14,21)?
- (c) [never/UK/2mk] LCM(14,21)?

T.II.5.001 · Unit fractions of amounts (UK wording) · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{6}{10}$.
- (b) [never/UK/2mk] Find $\frac{3}{5}$ of 55.
- (c) [never/UK/2mk] Calculate $\frac{3}{5} + \frac{2}{5}$.

T.II.5.002 · Non-unit fractions of amounts · challenge · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{2}{4}$.
- (b) [never/UK/2mk] Find $\frac{1}{2}$ of 12.
- (c) [never/UK/2mk] Calculate $\frac{1}{2} + \frac{1}{2}$.

T.II.5.003 · Equivalent fractions wall fluency · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Simplify $\frac{6}{16}$.
- (b) [never/UK/2mk] Find $\frac{3}{8}$ of 64.
- (c) [never/UK/2mk] Calculate $\frac{3}{8} + \frac{2}{3}$.

T.II.5.004 · Improper fractions ↔ mixed numbers fluency · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{2}{10}$.
- (b) [never/UK/2mk] Find $\frac{1}{5}$ of 40.
- (c) [never/UK/2mk] Calculate $\frac{1}{5} + \frac{4}{7}$.

T.II.5.005 · Fractions greater than 1 on number lines · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{6}{16}$.
- (b) [never/UK/2mk] Find $\frac{3}{8}$ of 56.
- (c) [never/UK/2mk] Calculate $\frac{3}{8} + \frac{1}{6}$.

T.II.5.006 · Fraction of a set with remainders context · challenge · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{6}{6}$.
- (b) [never/UK/2mk] Find $\frac{3}{3}$ of 12.
- (c) [never/UK/2mk] Calculate $1 + \frac{4}{3}$.

T.II.6.001 · Add/subtract fractions same denominator exam · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{4}{12}$.
- (b) [never/UK/2mk] Find $\frac{2}{6}$ of 42.
- (c) [never/UK/2mk] Calculate $\frac{1}{3} + \frac{1}{2}$.

T.II.6.002 · Add/subtract mixed numbers exam pack · core · UK_KS2,UK_KS3

- (a) [never/neutral/1mk] Simplify $\frac{8}{16}$.
- (b) [never/UK/2mk] Find $\frac{4}{8}$ of 56.
- (c) [never/UK/2mk] Calculate $\frac{1}{2} + \frac{3}{8}$.

T.II.6.003 · Multiply fractions by wholes and fractions · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{6}{18}$.
- (b) [never/UK/2mk] Find $\frac{3}{9}$ of 81.
- (c) [never/UK/2mk] Calculate $\frac{1}{3} + \frac{1}{4}$.

T.II.6.004 · Divide fractions by wholes (Y6) · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{10}{12}$.
- (b) [never/UK/2mk] Find $\frac{5}{6}$ of 72.
- (c) [never/UK/2mk] Calculate $\frac{5}{6} + \frac{2}{3}$.

T.II.6.005 · Fraction ↔ decimal ↔ percent early bridge · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{6}{12}$.
- (b) [never/UK/2mk] Find $\frac{3}{6}$ of 48.
- (c) [never/UK/2mk] Calculate $\frac{1}{2} + \frac{1}{6}$.

T.II.6.006 · Simplify fractions to lowest terms · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{2}{12}$.
- (b) [never/UK/2mk] Find $\frac{1}{6}$ of 36.
- (c) [never/UK/2mk] Calculate $\frac{1}{6} + \frac{1}{6}$.

T.II.6.007 · Fraction of a remaining amount multi-step · challenge · UK_KS2,UK_KS3

- (a) [never/neutral/1mk] Simplify $\frac{6}{4}$.
- (b) [never/UK/2mk] Find $\frac{3}{2}$ of 18.
- (c) [never/UK/2mk] Calculate $\frac{3}{2} + \frac{2}{3}$.

T.II.6.008 · Compare fractions with different denominators · core · UK_KS2

- (a) [never/neutral/1mk] Simplify $\frac{4}{14}$.
- (b) [never/UK/2mk] Find $\frac{2}{7}$ of 77.
- (c) [never/UK/2mk] Calculate $\frac{2}{7} + \frac{3}{5}$.

T.II.7.001 · Money calculations with £ and p (two decimals) · core · UK_KS2

- (a) [never/neutral/1mk] Total of £1.66 and £1.83?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.49. Change?
- (c) [never/UK/3mk] Buy three items £1.66, £1.83, £3.27 with a £10 note. Enough? Change?

T.II.7.002 · Rounding money to nearest pound/penny · core · UK_KS2

- (a) [never/neutral/1mk] Total of £2.50 and £0.64?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.14. Change?
- (c) [never/UK/3mk] Buy three items £2.50, £0.64, £1.49 with a £10 note. Enough? Change?

T.II.7.003 · Decimal ordering to 3 d.p. exam drills · core · UK_KS2

- (a) [never/neutral/1mk] Total of £1.13 and £2.60?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.73. Change?
- (c) [never/UK/3mk] Buy three items £1.13, £2.60, £2.28 with a £10 note. Enough? Change?

T.II.7.004 · Multiply/divide decimals by 10/100/1000 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £2.09 and £0.44?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £2.53. Change?
- (c) [never/UK/3mk] Buy three items £2.09, £0.44, £3.23 with a £10 note. Enough? Change?

T.II.7.005 · Thousandths place value (Y5–6) · core · UK_KS2

- (a) [never/neutral/1mk] Total of £1.29 and £2.69?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.98. Change?
- (c) [never/UK/3mk] Buy three items £1.29, £2.69, £3.38 with a £10 note. Enough? Change?

T.II.7.006 · Decimal × whole number money contexts · core · UK_KS2

- (a) [never/neutral/1mk] Total of £2.18 and £2.37?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £4.55. Change?
- (c) [never/UK/3mk] Buy three items £2.18, £2.37, £1.54 with a £10 note. Enough? Change?

T.II.7.007 · Decimal problems in measure contexts · core · UK_KS2

- (a) [never/neutral/1mk] Total of £0.95 and £2.26?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.21. Change?
- (c) [never/UK/3mk] Buy three items £0.95, £2.26, £1.47 with a £10 note. Enough? Change?

T.II.8.001 · Simple formulae in words then letters (Y6) · core · UK_KS2

- (a) [never/neutral/2mk] nth term of 2, 9, 16, 23, ...
- (b) [never/UK/1mk] 20th term: $a=2, d=7$.
- (c) [never/UK/2mk] S_n for $a=2, d=7$.

T.II.8.002 · Find pairs of numbers satisfying an equation · core · UK_KS2

- (a) [never/neutral/2mk] nth term of 4, 8, 12, 16, ...
- (b) [never/UK/1mk] 20th term: $a=4, d=4$.
- (c) [never/UK/2mk] S_n for $a=4, d=4$.

T.II.8.003 · Generate and describe linear sequences (Y6) · core · UK_KS2,UK_KS3

- (a) [never/neutral/2mk] nth term of 8, 12, 16, 20, ...
- (b) [never/UK/1mk] 20th term: $a=8, d=4$.
- (c) [never/UK/2mk] S_n for $a=8, d=4$.

T.II.8.004 · Use simple ratio language (Y6) · core · UK_KS2

- (a) [never/neutral/2mk] nth term of 3, 7, 11, 15, ...
- (b) [never/UK/1mk] 20th term: $a=3, d=4$.
- (c) [never/UK/2mk] S_n for $a=3, d=4$.

T.II.8.005 · Enumerate possibilities systematically (Y6) · challenge · UK_KS2

- (a) [never/neutral/2mk] nth term of 9, 16, 23, 30, ...
- (b) [never/UK/1mk] 20th term: $a=9, d=7$.
- (c) [never/UK/2mk] S_n for $a=9, d=7$.

T.II.9.001 · Imperial ↔ metric approximate equivalences · core · UK_KS2

- (a) [never/neutral/1mk] [Imperial ↔ metric approximate equivalences] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 6 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.II.9.002 · Convert between metric units fluently · core · UK_KS2

- (a) [never/neutral/1mk] [Convert between metric units fluently] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.II.9.003 · Area and perimeter of rectilinear shapes · core · UK_KS2

- (a) [never/neutral/1mk] [Area and perimeter of rectilinear shapes] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $6 \times 5 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 6) standing on its base.

T.II.9.004 · Volume of cubes and cuboids (cm^3) · core · UK_KS2

- (a) [never/neutral/1mk] [Volume of cubes and cuboids (cm^3)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.II.9.005 · Estimate angles; classify acute/obtuse/reflex · core · UK_KS2

- (a) [never/neutral/1mk] [Estimate angles; classify acute/obtuse/reflex] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.II.9.006 · Draw and measure angles to nearest degree · core · UK_KS2

- (a) [never/neutral/1mk] [Draw and measure angles to nearest degree] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 7 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 7) standing on its base.

T.II.9.007 · Translation parallel to axes only (Y5 constraint) · core · UK_KS2

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 46° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 46° ?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Translation parallel to axes only (Y5 constraint)»: build the figure, measure, verify the claimed relation.

T.II.9.008 · Reflection in lines parallel to axes · core · UK_KS2

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 69° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 69° ?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Reflection in lines parallel to axes»: build the figure, measure, verify the claimed relation.

T.II.9.009 · Coordinates in first quadrant plot & read · core · UK_KS2

- (a) [never/neutral/1mk] [Coordinates in first quadrant plot & read] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 5 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.II.9.010 · Four-quadrant coordinates intro (Y6) · core · UK_KS2

- (a) [never/neutral/1mk] [Four-quadrant coordinates intro (Y6)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 6 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.II.9.011 · Missing angles on a straight line / at a point · core · UK_KS2

- (a) [never/neutral/1mk] [Missing angles on a straight line / at a point] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 2 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.II.9.012 · Properties of regular polygons (sides/angles) · challenge · UK_KS2,UK_KS3

- (a) [never/neutral/1mk] [Properties of regular polygons (sides/angles)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 3 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.II.9.013 · Net of cubes and cuboids recognition · core · UK_KS2

- (a) [never/neutral/1mk] A cuboid measures $9 \times 2 \times 6$. How many faces does a correct net contain?
- (b) [never/UK/2mk] Which of these cannot fold to a cube: cross of 6 squares vs row of 6 squares? Explain.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Net of cubes and cuboids recognition»: build the figure, measure, verify the claimed relation.

T.II.9.014 · Identify 3D faces, edges, vertices · core · UK_KS2

- (a) [never/neutral/1mk] [Identify 3D faces, edges, vertices] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $6 \times 4 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 6) standing on its base.

T.II.9.015 · Area of triangles by counting half-squares · core · UK_KS2

- (a) [never/neutral/1mk] [Area of triangles by counting half-squares] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 5 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.II.9.016 · Scale factor enlarge by integer (Y6) · core · UK_KS2

- (a) [never/neutral/1mk] Lengths scale by 3. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement $SF=3$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Scale factor enlarge by integer (Y6)»: build the figure, measure, verify the claimed relation.

T.II.9.017 · Perimeter of polygons with missing sides · challenge · UK_KS2

- (a) [never/neutral/1mk] [Perimeter of polygons with missing sides] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 3 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 7) standing on its base.

T.II.9.018 · Draw circles with compass; label radius/diameter · core · UK_KS2

- (a) [never/neutral/1mk] [Draw circles with compass; label radius/diameter] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $6 \times 2 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Draw circles with compass; label radius/diameter»: build the figure, measure, verify the claimed relation.

T.II.9.019 · Time duration worded crossing midnight · challenge · UK_KS2

- (a) [never/neutral/1mk] [Time duration worded crossing midnight] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 2 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.II.9.020 · Convert hours–minutes–seconds · core · UK_KS2

- (a) [never/neutral/1mk] [Convert hours–minutes–seconds] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 6 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.II.9.021 · Find missing angle in triangle (Y6) · core · UK_KS2

- (a) [never/neutral/1mk] [Find missing angle in triangle (Y6)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 5 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.II.9.022 · Imperial-metric estimation set 1 · core · UK_KS2

- (a) [never/neutral/1mk] [Imperial-metric estimation set 1] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 5 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 7) standing on its base.

T.II.9.023 · Axis-parallel transform drill 1 · core · UK_KS2

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 58° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 58° ?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Axis-parallel transform drill 1»: build the figure, measure, verify the claimed relation.

T.II.9.024 · Imperial-metric estimation set 2 · core · UK_KS2

- (a) [never/neutral/1mk] [Imperial-metric estimation set 2] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 7 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.II.9.025 · Axis-parallel transform drill 2 · core · UK_KS2

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 50° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 50° ?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Axis-parallel transform drill 2»: build the figure, measure, verify the claimed relation.

T.II.10.001 · Pie charts: interpret simple sectors (Y6) · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 3, 8, 10, 16, 16, 17?
- (b) [allowed/UK/1mk] Range of 3, 8, 10, 16, 16, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 9. Four are 3, 8, 10, 16. Find the fifth.

T.II.10.002 · Calculate the mean (Y6 statutory) · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 5, 6, 9, 11, 15, 15?
- (b) [allowed/UK/1mk] Range of 5, 6, 9, 11, 15, 15?
- (c) [allowed/UK/2mk] Mean of five numbers is 8. Four are 5, 6, 9, 11. Find the fifth.

T.II.10.003 · Line graphs: interpret dual scales carefully · challenge · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 3, 10, 14, 14, 17, 17?
- (b) [allowed/UK/1mk] Range of 3, 10, 14, 14, 17, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 13. Four are 3, 10, 14, 14. Find the fifth.

T.II.10.004 · Timetables and dual bar charts combined · challenge · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 5, 6, 6, 8, 15, 16?
- (b) [allowed/UK/1mk] Range of 5, 6, 6, 8, 15, 16?
- (c) [allowed/UK/2mk] Mean of five numbers is 8. Four are 5, 6, 6, 8. Find the fifth.

T.II.10.005 · CCSS measurement data displays pack · core · US_CCSS_K5,UK_KS2

- (a) [allowed/neutral/1mk] Mean of 6, 9, 15, 16, 17, 18?
- (b) [allowed/US/1mk] Range of 6, 9, 15, 16, 17, 18?
- (c) [allowed/US/2mk] Mean of five numbers is 14. Four are 6, 9, 15, 16. Find the fifth.

T.II.10.006 · Mode and range early (UK secondary bridge) · core · UK_KS2,UK_KS3

- (a) [allowed/neutral/1mk] Mean of 5, 7, 11, 12, 15, 17?
- (b) [allowed/UK/1mk] Range of 5, 7, 11, 12, 15, 17?
- (c) [allowed/UK/2mk] Mean of five numbers is 8. Four are 5, 7, 11, 12. Find the fifth.

T.II.10.007 · Interpret misleading graphs critically · strange · UK_KS2,US_CCSS_68

- (a) [allowed/neutral/1mk] Mean of 8, 10, 10, 12, 13, 15?
- (b) [allowed/UK/1mk] Range of 8, 10, 10, 12, 13, 15?
- (c) [allowed/UK/2mk] Mean of five numbers is 13. Four are 8, 10, 10, 12. Find the fifth.

T.II.10.008 · Dual bar charts compare categories · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 4, 6, 16, 18, 18, 18?
- (b) [allowed/UK/1mk] Range of 4, 6, 16, 18, 18, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 14. Four are 4, 6, 16, 18. Find the fifth.

T.II.10.009 · Pie chart and mean combo 1 · challenge · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 5, 7, 8, 15, 16, 18?
- (b) [allowed/UK/1mk] Range of 5, 7, 8, 15, 16, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 12. Four are 5, 7, 8, 15. Find the fifth.

T.II.10.010 · Pie chart and mean combo 2 · challenge · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 7, 10, 11, 12, 16, 18?
- (b) [allowed/UK/1mk] Range of 7, 10, 11, 12, 16, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 7, 10, 11, 12. Find the fifth.

Answer Key — Era II

Brief solutions. Full reasoning in machine JSON.

T.II.1.001

- (a) LXVI — *Standard Roman mapping* → LXVI.
- (b) 80 — $LXXX = 80$.
- (c) LXVII — $66+1 = 67 \rightarrow LXVII$.

T.II.1.002

- (a) 1112 — *Add with carries*.
- (b) 1117 — *Column addition*.
- (c) 1112 — $762+350$.

T.II.1.003

- (a) 378 — *Exchange as needed*.
- (b) 373 — *Column method*.
- (c) 378 — $755-377$.

T.II.1.004

- (a) 863 — *Add with carries*.
- (b) 868 — *Column addition*.
- (c) 863 — $438+425$.

T.II.1.005

- (a) 954 — *Add with carries*.
- (b) 959 — *Column addition*.
- (c) 954 — $574+380$.

T.II.1.006

- (a) 806 — *Add with carries*.
- (b) 811 — *Column addition*.
- (c) 806 — $301+505$.

T.II.1.007

- (a) 865 — *Add with carries*.
- (b) 870 — *Column addition*.
- (c) 865 — $691+174$.

T.II.1.008

- (a) 994 — *Add with carries*.
- (b) 999 — *Column addition*.
- (c) 994 — $735+259$.

T.II.1.009

- (a) 1240 — *Add with carries*.
- (b) 1245 — *Column addition*.
- (c) 1240 — $617+623$.

T.II.1.010

- (a) 817 — *Add with carries*.
- (b) 822 — *Column addition*.
- (c) 817 — $426+391$.

T.II.1.011

- (a) 1407 — *Add with carries*.
- (b) 1412 — *Column addition*.
- (c) 1407 — $810+597$.

T.II.2.001

- (a) 1344 — *Long multiplication*.
- (b) 480 — *Place-value shift*.
- (c) 1344 — 48×28 .

T.II.2.002

- (a) 964 — *Add with carries*.
- (b) 969 — *Column addition*.
- (c) 964 — $573+391$.

T.II.2.003

- (a) 1175 — *Long multiplication*.

- (b) 470 — *Place-value shift.*
- (c) 1175 — 47×25 .

T.II.2.004

- (a) 224 — *Long multiplication.*
- (b) 140 — *Place-value shift.*
- (c) 224 — 14×16 .

T.II.2.005

- (a) 1189 — *Long multiplication.*
- (b) 410 — *Place-value shift.*
- (c) 1189 — 41×29 .

T.II.2.006

- (a) 630 — *Long multiplication.*
- (b) 420 — *Place-value shift.*
- (c) 630 — 42×15 .

T.II.2.007

- (a) 1034 — *Long multiplication.*
- (b) 470 — *Place-value shift.*
- (c) 1034 — 47×22 .

T.II.2.008

- (a) 675 — *Long multiplication.*
- (b) 450 — *Place-value shift.*
- (c) 675 — 45×15 .

T.II.2.009

- (a) 604 — *Add with carries.*
- (b) 609 — *Column addition.*
- (c) 604 — $287 + 317$.

T.II.2.010

- (a) 722 — *Add with carries.*
- (b) 727 — *Column addition.*
- (c) 722 — $465 + 257$.

T.II.2.011

- (a) 745 — *Add with carries.*
- (b) 750 — *Column addition.*
- (c) 745 — $303 + 442$.

T.II.3.001

- (a) 855 — *Add with carries.*
- (b) 860 — *Column addition.*
- (c) 855 — $505 + 350$.

T.II.3.002

- (a) 867 — *Add with carries.*
- (b) 872 — *Column addition.*
- (c) 867 — $284 + 583$.

T.II.3.003

- (a) 45 — *Exact division.*
- (b) 45 r 2 — *Bus-stop / short division.*
- (c) 45 each, remainder 2 — *Division with remainder.*

T.II.3.004

- (a) 648 — *Add with carries.*
- (b) 653 — *Column addition.*
- (c) 648 — $385 + 263$.

T.II.3.005

- (a) 39 — *Exact division.*
- (b) 39 r 7 — *Bus-stop / short division.*
- (c) 39 each, remainder 7 — *Division with remainder.*

T.II.3.006

- (a) 1221 — *Add with carries.*
- (b) 1226 — *Column addition.*
- (c) 1221 — $826 + 395$.

T.II.3.007

- (a) 41 — *Exact division.*
- (b) 41 r 7 — *Bus-stop / short division.*
- (c) 41 each, remainder 7 — *Division with remainder.*

T.II.3.008

- (a) 847 — *Add with carries.*
- (b) 852 — *Column addition.*
- (c) 847 — $600+247$.

T.II.3.009

- (a) 1155 — *Add with carries.*
- (b) 1160 — *Column addition.*
- (c) 1155 — $769+386$.

T.II.3.010

- (a) 650 — *Add with carries.*
- (b) 655 — *Column addition.*
- (c) 650 — $381+269$.

T.II.4.001

- (a) 1, 2, 4, 5, 10, 20 — *Pair listing.*
- (b) 10 — *Greatest common factor.*
- (c) 60 — xy/HCF .

T.II.4.002

- (a) 1, 2, 7, 14 — *Pair listing.*
- (b) 7 — *Greatest common factor.*
- (c) 42 — xy/HCF .

T.II.4.003

- (a) 1, 2, 4, 8, 16 — *Pair listing.*
- (b) 8 — *Greatest common factor.*
- (c) 48 — xy/HCF .

T.II.4.004

- (a) 1, 2, 7, 14 — *Pair listing.*
- (b) 7 — *Greatest common factor.*
- (c) 42 — xy/HCF .

T.II.5.001

- (a) $3/5$ — *Divide by gcd.*
- (b) 33 — $55\div5\times3$.
- (c) 1 — *Common denominator.*

T.II.5.002

- (a) $1/2$ — *Divide by gcd.*
- (b) 6 — $12\div2\times1$.
- (c) 1 — *Common denominator.*

T.II.5.003

- (a) $3/8$ — *Divide by gcd.*
- (b) 24 — $64\div8\times3$.
- (c) $25/24$ — *Common denominator.*

T.II.5.004

- (a) $1/5$ — *Divide by gcd.*
- (b) 8 — $40\div5\times1$.
- (c) $27/35$ — *Common denominator.*

T.II.5.005

- (a) $3/8$ — *Divide by gcd.*
- (b) 21 — $56\div8\times3$.
- (c) $13/24$ — *Common denominator.*

T.II.5.006

- (a) 1 — *Divide by gcd.*
- (b) 12 — $12\div3\times3$.
- (c) $7/3$ — *Common denominator.*

T.II.6.001

- (a) $1/3$ — Divide by gcd.
- (b) 14 — $42 \div 6 \times 2$.
- (c) $5/6$ — Common denominator.

T.II.6.002

- (a) $1/2$ — Divide by gcd.
- (b) 28 — $56 \div 8 \times 4$.
- (c) $7/8$ — Common denominator.

T.II.6.003

- (a) $1/3$ — Divide by gcd.
- (b) 27 — $81 \div 9 \times 3$.
- (c) $7/12$ — Common denominator.

T.II.6.004

- (a) $5/6$ — Divide by gcd.
- (b) 60 — $72 \div 6 \times 5$.
- (c) $3/2$ — Common denominator.

T.II.6.005

- (a) $1/2$ — Divide by gcd.
- (b) 24 — $48 \div 6 \times 3$.
- (c) $2/3$ — Common denominator.

T.II.6.006

- (a) $1/6$ — Divide by gcd.
- (b) 6 — $36 \div 6 \times 1$.
- (c) $1/3$ — Common denominator.

T.II.6.007

- (a) $3/2$ — Divide by gcd.
- (b) 27 — $18 \div 2 \times 3$.
- (c) $13/6$ — Common denominator.

T.II.6.008

- (a) $2/7$ — Divide by gcd.
- (b) 22 — $77 \div 7 \times 2$.
- (c) $31/35$ — Common denominator.

T.II.7.001

- (a) £3.49 — Add pence; convert.
- (b) £1.51 — Subtract.
- (c) Yes, £3.24 — Compare sum with 1000p.

T.II.7.002

- (a) £3.14 — Add pence; convert.
- (b) £1.86 — Subtract.
- (c) Yes, £5.37 — Compare sum with 1000p.

T.II.7.003

- (a) £3.73 — Add pence; convert.
- (b) £1.27 — Subtract.
- (c) Yes, £3.99 — Compare sum with 1000p.

T.II.7.004

- (a) £2.53 — Add pence; convert.
- (b) £2.47 — Subtract.
- (c) Yes, £4.24 — Compare sum with 1000p.

T.II.7.005

- (a) £3.98 — Add pence; convert.
- (b) £1.02 — Subtract.
- (c) Yes, £2.64 — Compare sum with 1000p.

T.II.7.006

- (a) £4.55 — Add pence; convert.
- (b) £0.45 — Subtract.
- (c) Yes, £3.91 — Compare sum with 1000p.

T.II.7.007

- (a) £3.21 — Add pence; convert.
- (b) £1.79 — Subtract.

(c) Yes, £5.32 — Compare sum with 1000p.

T.II.8.001

(a) $7n + -5$ — Difference 7.

(b) $135 - a + (n-1)d$.

(c) $335 - n/2[2a + (n-1)d]$.

T.II.8.002

(a) $4n$ — Difference 4.

(b) $80 - a + (n-1)d$.

(c) $220 - n/2[2a + (n-1)d]$.

T.II.8.003

(a) $4n + 4$ — Difference 4.

(b) $84 - a + (n-1)d$.

(c) $260 - n/2[2a + (n-1)d]$.

T.II.8.004

(a) $4n + -1$ — Difference 4.

(b) $79 - a + (n-1)d$.

(c) $210 - n/2[2a + (n-1)d]$.

T.II.8.005

(a) $7n + 2$ — Difference 7.

(b) $142 - a + (n-1)d$.

(c) $405 - n/2[2a + (n-1)d]$.

T.II.9.001

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.002

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.003

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.004

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.005

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.006

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.007

(a) 46° — Corresponding equal.

(b) 134° — Sum 180° .

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.II.9.008

(a) 69° — Corresponding equal.

(b) 111° — Sum 180° .

(c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.II.9.009

(a) Top view (from above) — Plan = bird's-eye.

(b) 6 — Six faces.

(c) Isosceles triangle with base L — Front sees triangular profile.

T.II.9.010

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.011

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.012

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.013

- (a) 6 — *One face per side.*
- (b) Row of 6 cannot (exceeds max chain length 4) — *Cube-net rules.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.II.9.014

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.015

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.016

- (a) 9 — SF^2 .
- (b) 18 cm — $\times SF$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.II.9.017

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.018

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.II.9.019

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.020

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.021

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.022

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.023

- (a) 58° — *Corresponding equal.*
- (b) 122° — *Sum 180° .*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.II.9.024

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.II.9.025

- (a) 50° — *Corresponding equal.*
- (b) 130° — *Sum 180° .*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.II.10.001

- (a) 11.67 — *Sum $70 \div 6$.*
- (b) 14 — *max–min.*
- (c) 8 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.002

- (a) 10.17 — *Sum $61 \div 6$.*
- (b) 10 — *max–min.*
- (c) 9 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.003

- (a) 12.5 — *Sum $75 \div 6$.*
- (b) 14 — *max–min.*
- (c) 24 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.004

- (a) 9.33 — *Sum $56 \div 6$.*
- (b) 11 — *max–min.*
- (c) 15 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.005

- (a) 13.5 — *Sum $81 \div 6$.*
- (b) 12 — *max–min.*
- (c) 24 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.006

- (a) 11.17 — *Sum $67 \div 6$.*
- (b) 12 — *max–min.*
- (c) 5 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.007

- (a) 11.33 — *Sum $68 \div 6$.*
- (b) 7 — *max–min.*
- (c) 25 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.008

- (a) 13.33 — *Sum $80 \div 6$.*
- (b) 14 — *max–min.*
- (c) 26 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.009

- (a) 11.5 — *Sum $69 \div 6$.*
- (b) 13 — *max–min.*
- (c) 25 — $5 \times \text{mean} - \text{sum of four}$.

T.II.10.010

- (a) 12.33 — *Sum $74 \div 6$.*
- (b) 11 — *max–min.*
- (c) 15 — $5 \times \text{mean} - \text{sum of four}$.